

Teach Project Management, Pack an Agile Punch

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Abstract

Agile software development methods have begun to gain wide acceptance by the global software industry. However, project managers with academic training in ASD are a rarity. This paper outlines a course in IS Project Management that is well-suited for an audience consisting of students from the information systems, business and computer science streams, who normally make up a graduate class. The course begins by adopting a process-neutral perspective towards activities such as requirements and estimation, before proceeding to discuss Agile methods. A team project allows participants to develop a hands-on feel for ASD, with the help of an Agile PM platform. The author has recently taught the course as proposed at his institution. The paper makes some observations on the effect of prior work experience on team performance. It concludes with some testimonials from professional participants, on the impact that the course has had on their practices at work.

1. Introduction

It is well known that the project manager plays a pivotal role in steering software missions. In an extensive survey [1], Verner and Evanco examine the influence of key aspects such as requirements management, cost-effort estimation, and risk assessment on the final outcome of a project. Managers who are ill-groomed in these areas may go on to script veritable disasters, as illustrated by Robert Glass in his famous collection of “runaways” [2].

Academic institutions continuously redesign their programs to better prepare the students for professional careers. The IS 2010 Report [3] has suggested including a course in IS Project Management (abbreviated henceforth as ISPM) within the core of an undergraduate computer science curriculum. An advanced version of this course is offered as an elective to graduate students at many universities – the reader can consult [4] for a good survey.

The PMBOK framework [5] classifies project activities into planning, scoping, estimating time and cost, scheduling, ensuring quality, team building, communicating, managing risks, and overseeing the contract. While this categorization applies well to the context of software projects, any instructor on the subject will have trouble discussing such a panoply of topics within the short duration of a 10-week quarter or even a 15-week semester.

One way to address this problem of plenty is to draw from the principles of project-based learning [6] and experiential learning [7]. Because the topics warrant a considerable amount of exposition by the instructor, neither learning approach would be applicable in its entirety. For instance, it would be difficult for students to get system requirements right on their own, without the instructor covering the subject in adequate detail.

Layman *et al* [8] describe a two-stage, lab-driven course in software engineering, which is targeted at CS undergraduates. After exploring the many nuances of software development in a process-neutral manner, the students team up to implement a project in a methodology of their choice. Activities are designed to cater to the needs of a tech-savvy and collaborative “Millennial” generation, namely, those born after 1982.

A graduate course in ISPM would be vastly different from its undergraduate counterpart. The audience typically consists of students from multiple streams: information systems, business and computer science. Most would have worked for a few years in IT, and may not require education on “downstream” aspects such as design and coding. A few may even be enrolled in a continuing education program, such as a weekend MBA.

Over the last decade, Agile software development methodologies (ASD) like Scrum have steadily penetrated the software industry [9], in collocated and distributed project settings. Even though ASD is attractive in its packaging, Barry Boehm argues [10] that planned and agile methodologies each have a “home ground” of projects, where they would perform well. A judicious mix of methods from the two approaches would work more universally.

There are no examples in literature of an ISPM course for a heterogeneous graduate audience as described, which also includes an Agile project. The paper addresses this lacuna. It suggests some web-based platforms for facilitating course and project administration. Based on the author’s experiences with delivering the course, some observations are made regarding the effect of prior work experience on a student’s performance in project activities.

2. Course design

As highlighted earlier, teaching a graduate ISPM course involves a delicate balancing act. Because the course is usually offered as an elective, enrollment is voluntary, and participants tend to be motivated learners. However, the instructor has to cover a wide range of topics within a short timeframe. The following sections propose a design for an ISPM course that can cater to these competing needs, while remaining contemporary.

2.1. Pedagogy

Taking the cue from Layman *et al* [8], the proposed course is delivered in two stages. Instructor-led discussions form a large component of classroom discussions. They are based on pre-assigned seminal readings from journal publications. A team project, which is announced during the second half, serves as a vehicle for experiential and collaborative learning around Agile methods. The minimum suggested course duration is 10 weeks.

From time to time, students take part in structured soft-skill exercises that reinforce the communications aspect, which is lacking in many an ISPM course, according to Du *et al* [4]. A project manager is expected to spend a lot of time in meetings, and will therefore gain from training in facilitation. Because the PM is the first point of contact for a project’s stakeholders on any issue, conflict resolution and negotiation skills are essential.

Participants are assessed on the basis of their performance in a variety of tasks. They are expected to have had some exposure to the software industry, preferably for the duration of three or more years. Consequently, points are awarded for class participation. Their work experience also comes in handy on the project, as will be demonstrated later. Quizzes, a final and the project deliverables constitute the remaining graded components.

2.2. Process-neutral coverage

Regardless of the methodology adopted by a software team to execute a project, there are a few universal truths. Fred Brooks’ insights on managing scope in *The Mythical Man Month* are timeless. Similarly, Tom DeMarco and Tim Lister derive explicit linkages between properly motivated teams and productivity in *Peopleware*. The instructor could then proceed to discuss more general topics such as risk management, methodologies, and so on.

The IS Curriculum Report [3] and the PMBOK [7] offer excellent guidelines to structure the topics for the proposed ISPM course. Table 1 maps the knowledge areas of PMBOK to seminal readings in a software context. These readings are by no means comprehensive. However, when combined with the real-world experiences of the participants, each has the potential to generate meaningful discussions within the classroom.

Based on an extensive survey, Du *et al* [4] point out some deficiencies in ISPM course offerings. Notably, there is a reduced emphasis on critical aspects of project management such as procurement and outsourcing. These topics are relevant to today's globalised context. The proposed course attempts to bridge these gaps.

Table 1: PMBOK Topic Mappings

Knowledge area on the Project	Suggested readings and books
Integration Management	Goldstein, H. <i>Who Killed the Virtual Case File?</i> IEEE Spectrum (2005)
Scope Management	Fairley, R.E. & Wilshire, M.J. <i>Why the Vasa Sank? 10 Problems and Some Antidotes for Software Projects</i> IEEE Software (2003)
Time Management	Brooks, F. <i>The Mythical Man Month: Essays on Software Engineering</i> , 2 nd Edition (1995)
Cost Management	Jones, C. <i>Activity Based Software Costing</i> . Computer (1996)
Quality Management	Whittaker, J.A. <i>What is Software Testing? And Why is it So Hard?</i> IEEE Software (2000)
HR Management	De Marco, T. & Lister, T. <i>Peopleware: Productive Projects and Teams</i> , 2 nd Edition. Dorset House (1999)
Communications Management	Teasley, S.D. et al. <i>Rapid Software Development through Team Collocation</i> . IEEE Transactions on Software Engineering (2002)
Risk Management	Boehm, B. <i>Software Risk Management: Principles and Practices</i> IEEE Software (1991)
Procurement Management	Lacity, M. <i>Lessons in Global Information Technology Outsourcing</i> . Computer (2002)

A textbook can supplement those topics that are not covered by the prescribed readings. Kathy Schwalbe's *Information Technology Project Management* is an excellent choice for students who are interested in a PMBOK-driven treatment of the subject. For a practitioner-oriented treatment, there is no better reference than Steve McConnell's *Rapid Development*, whose caselets are quite realistic.

While the PMBOK classification is broadly applicable, the setting of software demands domain-specific coverage of risk management, estimation techniques, quality measures, and so on. Cost and effort estimation is the chief area of difficulty for software professionals. There are scientific approaches (e.g. COCOMO II, function points) to tackle this problem. The instructor may prescribe additional readings and exercises to cover these aspects.

2.3. Agile emphasis

A discussion of the Unified Process (see Larman [11]) is an effective vehicle to transition to Agile software development methods. The Agile Manifesto and Scrum Guide [12] are effective catalysts for a lively discussion in the classroom. Instructors must emphasize the fact that planning is incorporated into an Agile process at multiple junctures, such as at the iteration and sprint levels.

Participants can now re-examine aspects such as requirements and scope management with an Agile lens. The instructor can contrast the construct of a user story [13] with a use case, and explain its metamorphosis during the Agile lifecycle. The product backlog is an effective scope management tool, enabling the team to respond to the changing needs of the customer. Quality is ensured by pair programming, continuous integration and so on.

The technique of estimating with story points can be contrasted with the function point approach. The instructor may choose to conduct a Planning Poker exercise, asking the teams to size the many tasks they have drawn up on their project, and arrive at consensus estimates. There is a wealth of audiovisual material in the public domain on the various aspects of Agile, by exponents such as Ken Schwaber and Mike Cohn.

3. Course realization

For many years, the author has been teaching a 30-hour, 20-session ISPM course with a software emphasis, at the Indian Institute of Management Bangalore (IIMB), one of India's leading business schools. The course is titled Managing Software Projects (MSP). The most recent batch of students (the course concluded in September 2010) form the subject of the discussions that follow.

3.1. Student profile

27 participants registered for MSP 2010 (see Figure 1). Two-thirds had relatively lower experience in IT (averaging 3.8 years), and were enrolled full-time in the MBA program. The remaining third had significantly higher work experience (8.5 years), and were enrolled in a continuing program for IT executives. We shall refer to these executives as EMBA's. About one thirds of the participants had already been exposed to Agile methods at work.

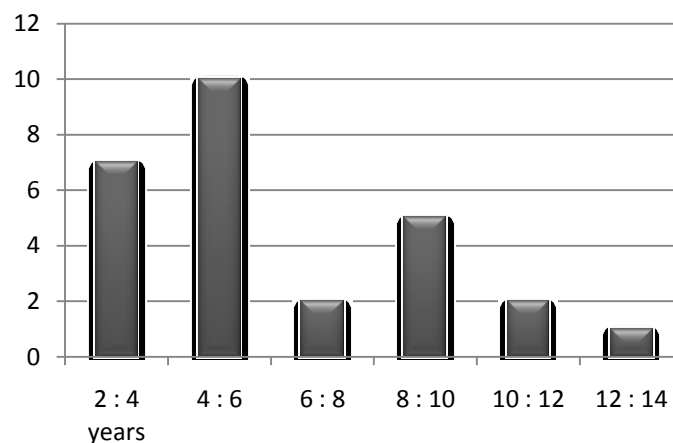


Figure 1: Experience profile of the participants

3.2. Project Mandate

After exploring the basics of Agile, the participants were introduced to their project halfway through the course. The assignment was broken into three phases (consult Appendix A for details), to be completed within five weeks by a four-member team. Given the practitioner background of the participants and the business school setting, no coding was involved. Instead, the final project deliverable was a GUI prototype.

Project teams were asked to conceptualize an application to facilitate Planning Poker activities online, for use on distributed projects. Because the instructor had elaborated on the Delphi method in class, the teams had a fair idea about their assignment.

3.3. Technology mediation

A course of this nature exacts a hefty price: automation is a savior at each step. An LMS or learning management system lessens the administrative burden on instructors by enabling them to display documents and hyperlinks, conduct quizzes online and grade assignments. Students can in turn upload deliverables, and create threads to raise their doubts, so their experienced peers can resolve them. The author's institution hosts an instance of Moodle.

In addition to an LMS, the participants used special-purpose software to accomplish their project objectives. Mingle[®] by Thoughtworks is a PM platform, which is designed to handle agile constructs such as user stories, story walls, sprints, and so on. For their prototyping needs, the teams made use of an open source tool called the Pencil Project, which helped them to quickly mock up screens and even navigate through them.

4. Project outcomes

The class of 27 participants grouped together into six teams of 4 members each and one team with 3 members. A structured brainstorm conducted in class helped prepare the participants for their requirements phase. The proceedings were captured on video and posted online for the participants to watch independently. The learnings from this exercise were put to use during Phase I, while arriving at the application's features.

The teams differed in the quality of detail within their user stories. Table 2 contrasts the level of specificity in user story descriptions between two teams. There was a wide variance in the number of stories (features) that different teams had conceived. It could be safely concluded that teams that uploaded a larger number of stories had deliberated their mission in greater depth.

Table 2: Team differences

Example Story from Team 1	Example Story from Team 2
As a Product Owner, I would like to register a team of members, so that I can initiate a Delphi Estimation Exercise.	As a moderator, I should be able to tag multiple estimators to a project.

During Phase II, the teams were asked to arrive at a consensus on their story point estimates via a Planning Poker exercise. Teams with a smaller set of stories had trouble converging on their estimates. Usually, they discovered some stories that were epics, which had to be broken down further. Those with fine-grained stories did better at this task, but had problems completing the exercise within the session time of 90 minutes.

Phase III immersed the teams in prototyping tasks with the help of the Pencil Project. A participant volunteer, who had mastered the tool, made a presentation to the class. Because students in a business school tend to be competitive, the teams went on to prototype all the stories. The author had only expected for them to work on the first couple of sprints. As an upshot, it can be concluded that the project load was fair.

The author periodically checked upon the activities of the project teams on Mingle. While some teams showed an initial hesitation to engage with the tool, they became more adventurous once they grew familiar with it. For instance, one team published a status pie-chart, even though they were not expected to do so. Barring one instance, all of the teams were able to split their stories into well-balanced sprints.

A couple of teams toyed with the concept of transitions, which prevent a user from inadvertently moving a story card to an invalid stage (e.g. from Requirements to Testing without Development). This flavor of exploratory learning was manifest at different levels across all the teams, and throughout the project. Naturally, there was a wide variance in the composition and the quality of the final deliverables.

Several teams exploited Mingle's reporting features to generate informative visuals, burn-down charts, release summaries, and so on (see Appendix B). While some time could be dedicated by the instructor in class to explain the idea of a burn-down chart, there is simply no substitute for the understanding that a team derives, by comparing its progress against that predicted by the tool.

As a final step, the instructor scheduled face-to-face sessions with each project team. Most had implemented role-based logins, which enabled different functions, based on whether the user was a PM or a group member. A thoughtful few could trace a screen back to the relevant story, demonstrating their appreciation for the process-neutral aspect of traceability.

5. Performance analysis

Many practitioners strongly believe (see [6] for example) that agile project efforts are best suited for experienced teams of professionals, who are all "above average." Given the diversity of backgrounds within the MSP participants, it would be worthwhile to put this assumption to test, and derive conclusions. Given the small population size, it would be necessary to invoke non-parametric tests to make our assertions.

Because EMBA students, the group with greater work experience, were engaged full-time within the IT industry, it would be incorrect to make performance comparisons with their MBA peers, along dimensions like the final grade. However, the EMBA students had grouped to form project teams. Therefore, we can make a fair comparison with MBA students along the various project components. Phases I and II carried 10 points each, whereas Phase III accounted for 15 points.

Hypothesis 1: There is no relationship between the number of years of prior IT experience and the scores on the project.

As illustrated by Table 3, a Spearman's Rank Correlation Test on IT Experience versus the overall Project Score yielded a rho value of 0.254, which may not be significant. However, there was a significant positive correlation of 0.483 (level < 0.05) between IT Experience and the score for Phase I. This indicates that one's proficiency with project requirements increases with professional experience.

Table 3: Bi-variate correlations for various project dimensions

		ITExp	Del 1	Del 2	Del 3	PScore
ITExperience	Correlation Coefficient	1.000	.483*	.037	-.217	.254
	Sig. (2-tailed)	.	.011	.853	.277	.202
	N	27	27	27	27	27
Deliverable1	Correlation Coefficient	.483*	1.000	-.108	-.240	.434*
	Sig. (2-tailed)	.011	.	.590	.228	.024
	N	27	27	27	27	27
Deliverable2	Correlation Coefficient	.037	-.108	1.000	.463*	.765**
	Sig. (2-tailed)	.853	.590	.	.015	.000
	N	27	27	27	27	27
Deliverable3	Correlation Coefficient	-.217	-.240	.463*	1.000	.540**
	Sig. (2-tailed)	.277	.228	.015	.	.004
	N	27	27	27	27	27
ProjectScore	Correlation Coefficient	.254	.434*	.765**	.540**	1.000
	Sig. (2-tailed)	.202	.024	.000	.004	.
	N	27	27	27	27	27

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Notice the negative correlation between IT Experience and the Phase III score, which in turn is positively correlated with the overall Project Score. The increased weightage for Phase III could explain the reduced dependency of the overall Project Score on IT Experience.

Given that the full-time MBA students were residing in campus dormitories, they grouped together and were able to work more closely on the “implementational aspects” of the project, such as uploading user stories and creating user interfaces. This fact was verified by the accounts of the teams. The EMBA students were not only spread across the city, but also had to attend to work duties during much of the week.

Hypothesis 2: *There is no difference between the two groups of students (i.e. MBA and EMBA) regarding the overall project scores.*

As evidenced with a Mann-Whitney test (see Table 4), the hypothesis stays. However, this result could be specific to the mixed profile of the participants.

Table 4: Hypothesis Test Summary

Null Hypothesis	Test	Significance	Decision
The distribution of ProjectScore is the same across categories of Program	Independent Samples Mann-Whitney U Test	0.067	Retain the null hypothesis

Asymptotic significances are displayed. The significance level is 0.05

6. Testimonials

The course was received well by the participants, judging from their final feedback. Upon soliciting their reactions, the author received the responses listed below:

As a QA Manager, I am able to apply many principles in my day to day activities, especially the concepts of Evidence Based Scheduling, Unified Process, and the Agile approach. I am responsible for ensuring the quality of the software released. Having gone through this course, I now enjoy an advantage while negotiating over user stories and estimating story points. This approach has helped me have a "Clear Say" in Scrum discussions on items that may be included in the release, rather than committing first and then taking them off the plan later. (EMBA)

This was one of the most useful courses for me in the MBA program. The synthesis of risk management, requirements gathering, quality control and most importantly the agile methodology of project management has given me a strong foundation to execute a software project for the Career Development Cell, to facilitate its placement process. In fact, the experience of the project we did in the course gave me the confidence to start using these principles immediately. I could see how my work transformed into a more organized and measurable piece. Many of my peers were impressed by the planning, and thought it had taken me a considerable amount of time. The truth, however, was just the opposite: having been exposed to agile methods in my academic project, reusing the concepts at work was a breeze. (MBA)

Topics such as effort estimation, person days analysis, use case modeling (casual and fully dressed) are still helping me create value for the customer by streamlining various activities. The overall knowledge I gained has helped me use Quantitative PM techniques for effort estimation/schedule slippage calculations, etc. I am able to adapt a modified standup meeting in my current project. Although we don't use the chicken/pig concept, the agenda of this standup in Scrum has helped me to focus on key points worth a discussion. Overall it helped me prioritize what is important from what isn't during standups. (EMBA)

We shifted from a 1-month Scrum sprint to a 1 week sprint as an experiment, after studying the theory behind the method, and exploring ways to tweak the process to our team's needs. The weekly scrum frequency and stand-ups have given us a sense of urgency to complete our stories faster. One month used to be a long timeframe. Developers tended to relax at the beginning of the month and pull all-nighters later. A process where everybody is charged to deliver each week (Monday to Friday) has eliminated instances where folks need to work over weekends to cover up. (EMBA)

7. Conclusion

This paper outlines a course in IS Project Management, which can be taught to a heterogeneous mix of graduate students, with a substantial Agile project component. Many project deliverables have shown evidence of the application of process-neutral concepts like traceability.

It emerges that participants with higher levels of industry experience fare better with documenting requirements, which is important to the success of any software mission. It is expected that the course as proposed would be equally effective when the project carries a different mandate.

8. Acknowledgements

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Appendix A: Project Phase Description

Phase I	- Through a structured brainstorm, generate features of the product you will build. Express them as stories and tasks. Make sure they follow the INVEST & SMART guidelines. - Create a project in Mingle and load your stories	1 week
Phase II	- Rank the stories based on the information on functional priority - NOT by the effort it will take to deliver. For each story/task, arrive at independent estimates as team members. - Undertake an Estimation Poker exercise in class, recording the results on a spreadsheet. Capture the risks involved and list your assumptions. - Break the project into sprints, assuming a team of 3 members plus a ScrumMaster. - Update your Mingle project instance with all the details. Flesh out 5 Pencil screens at various levels of complexity	2 weeks
Phase III	- Use Pencil to flesh out screens for your product. Address as many sprints as possible. - Publish your screens periodically online via a website that you will be provided. - Update your Mingle instance periodically	2 weeks

Appendix B: Progress Reports

